Analog Synthesizer

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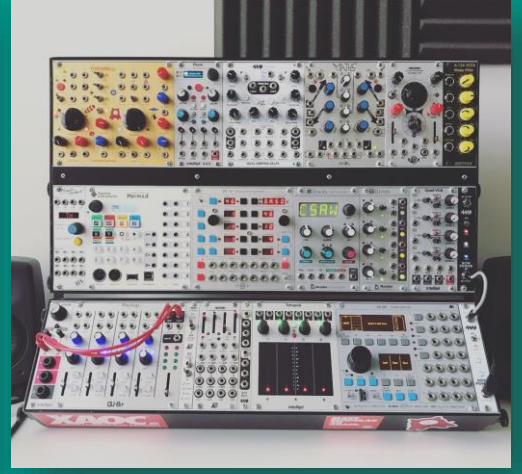
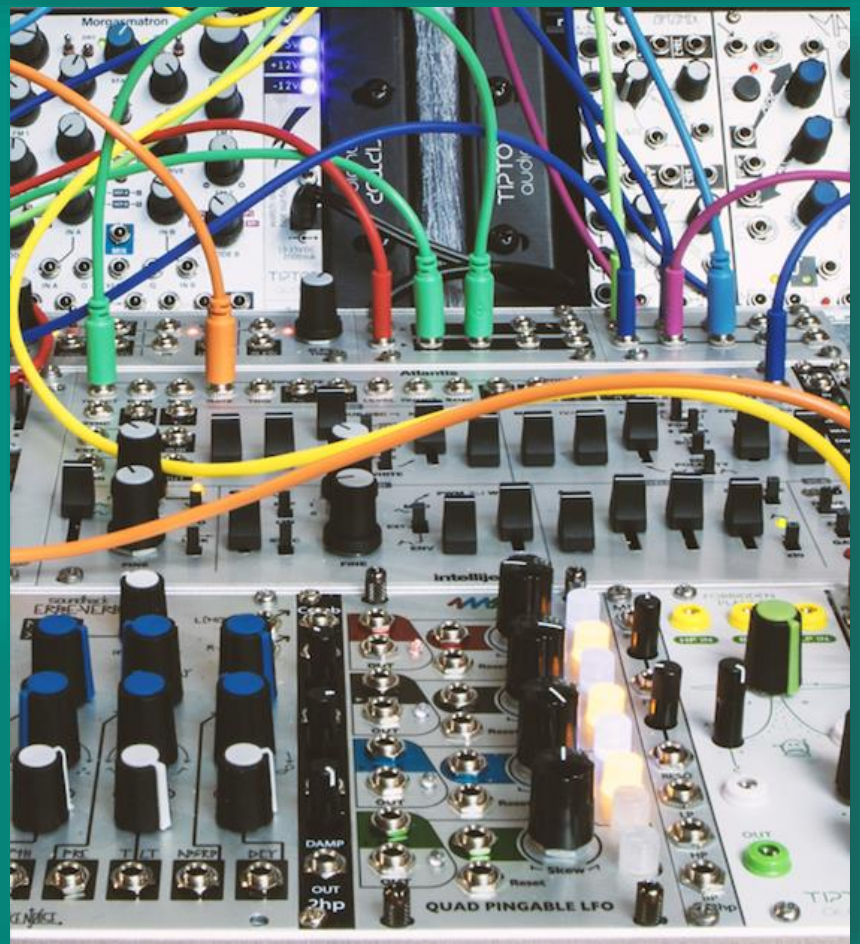
Introduction





- Shapes
- Forms
- Warmth
- Crafty
- Sonic

My Choice & Objective

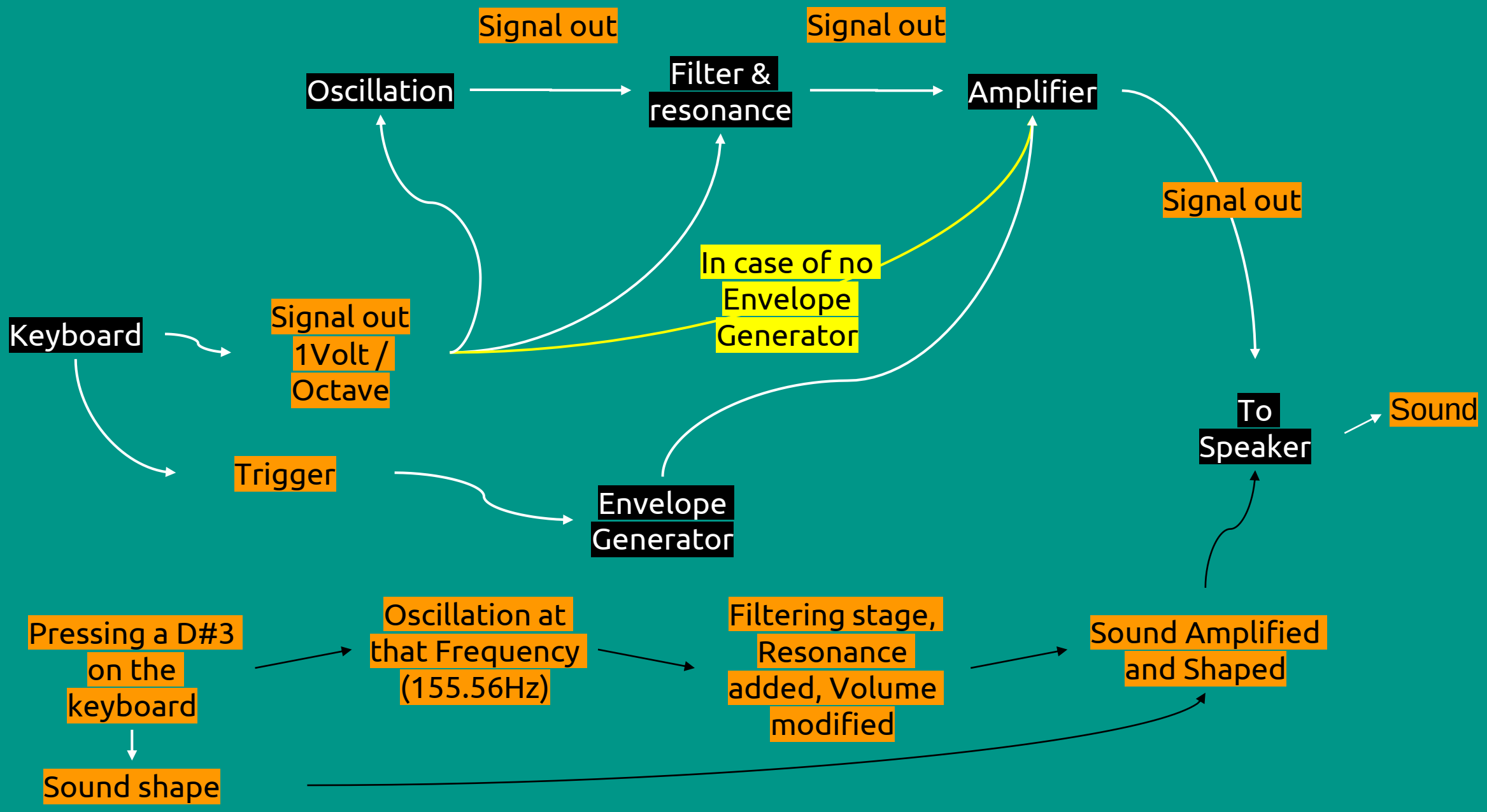
Why I Chose Analog Modular

- Strong preference for the analog sound.
- The ultimate "build-your-own-instrument" experience.
- Allows for infinite customization.
- Easy future upgrades and improvements.

Objective

- A 6-waveform analog synthesizer
- Voltage-controlled operation.
- Includes a frequency modulation input.
- Features a filter with a roll-off rate of 12 dB/octave.
- Stable output signal.

Block Diagram


Signal out
Signal out
Oscillation
Filter & resonance
Amplifier
Signal out
In case of no Envelope Generator
Keyboard
Signal out 1Volt / Octave
To Speaker
Sound
Trigger
Envelope Generator
Pressing a D#3 on the keyboard
Oscillation at that Frequency (155.56Hz)
Filtering stage, Resonance added, Volume modified
Sound Amplified and Shaped
Sound shape

System Design

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The project was divided into six key phases:

1 - Controller
2 - Voltage-Controlled Oscillator
3 - Voltage-Controlled Filter
4 - Voltage-Controlled Amplifier
5 - Envelope Generator
6 - Power Supply

Controller

Control voltage ranging from 0V to 5V.

Three independent outputs to control each module individually.

- Pitch For the VCO
- Gate For the VCF
- Mod for the VCA/EG

Control Voltage Section

Pitch → VCO

Gate → VCF

Mod → VCA

Voltage-Controlled Oscillator

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Coarse to adjusts the pitch of the oscillator in Larger increments

Fine to adjusts the pitch of the oscillator in smaller increments

Frequency Modulation input

Adjust the Frequency Modulation Level

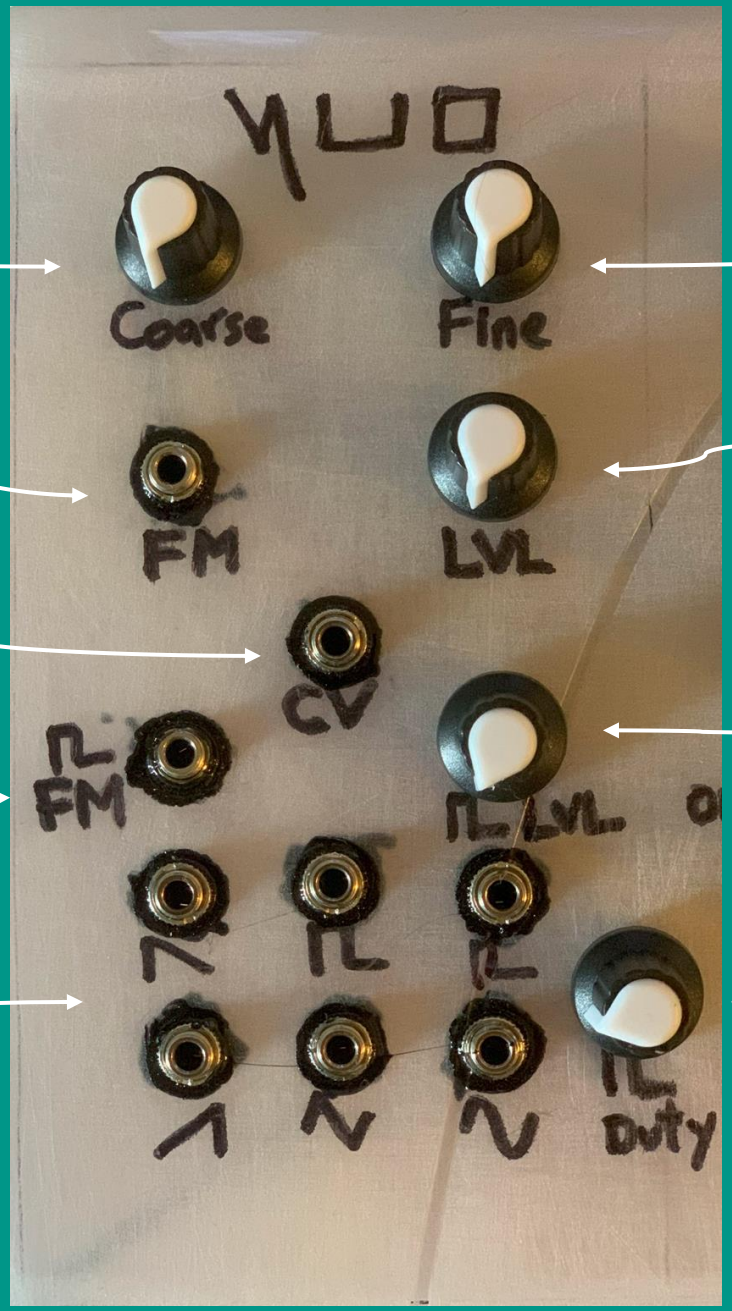
Control Voltage input

Pulse Width Frequency Modulation Level

Pulse wave Frequency Modulation input

Pulse Width Duty Cycle

6 Waveforms (Sawtooth, Square, Pulse, Reverse Saw, Triangle and sine)



Voltage-Controlled Filter

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Cutoff Frequency Knob

Resonance Knob

Input signal
From VCO

Adjust The Input Volume

Control Voltage input

Adjust the Control voltage
Level

Type of filter Selection
Low pass
Band pass
High pass

Output Signal

Voltage-Controlled Amplifier

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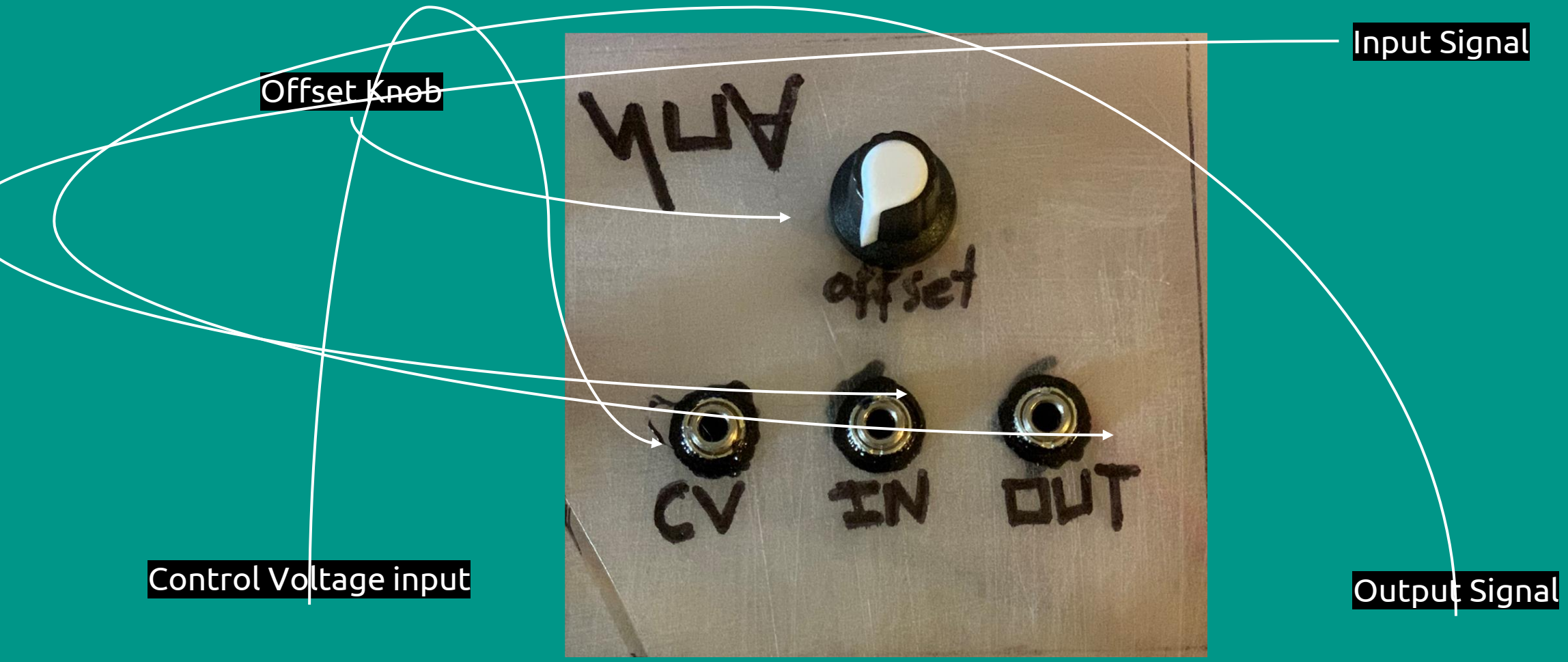

Input Signal
Offset Knob
offset
CV
IN
OUT
Control Voltage input
Output Signal

Envelope Generator

Attack Knob

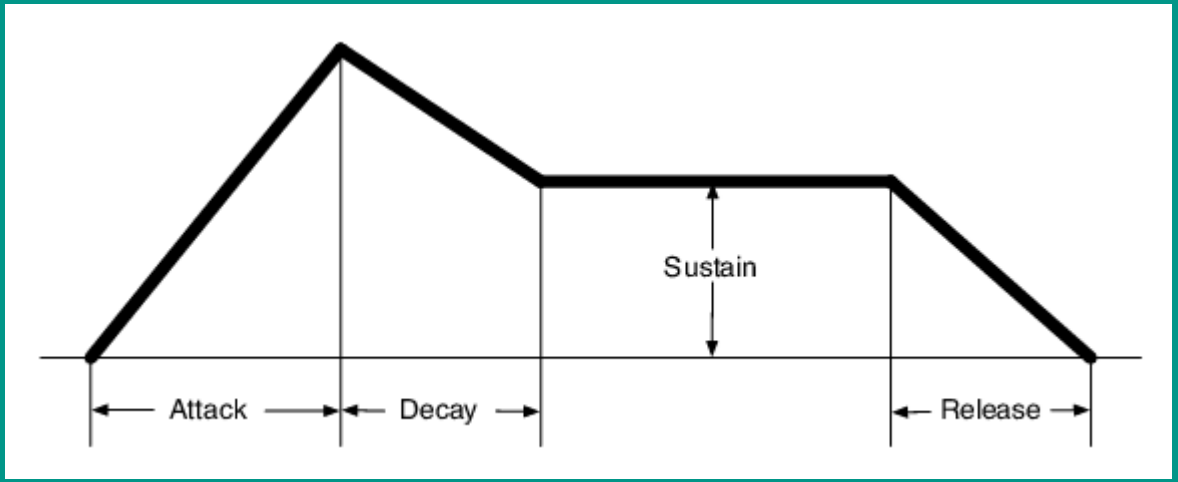
Decay Knob

Sustain Knob

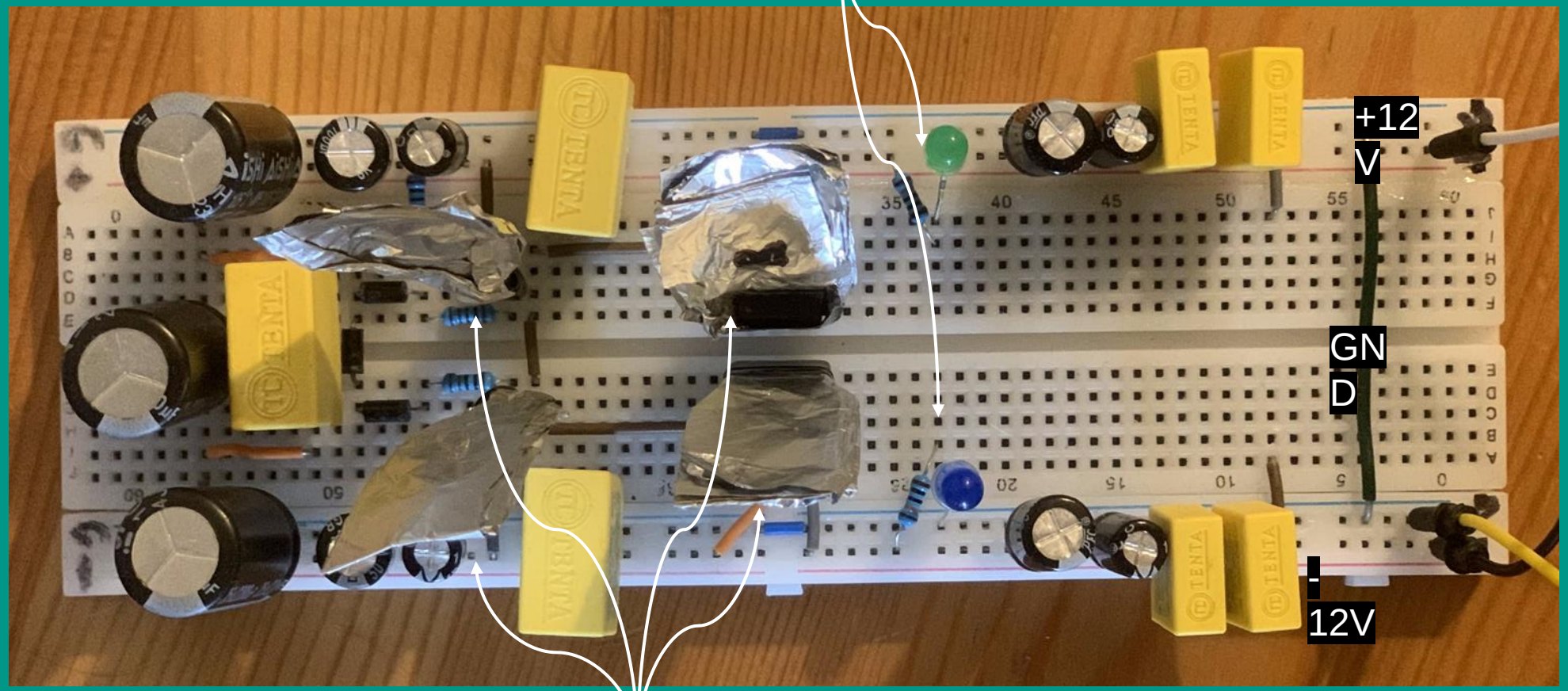
Release Knob

Trigger Input
From Keyboard

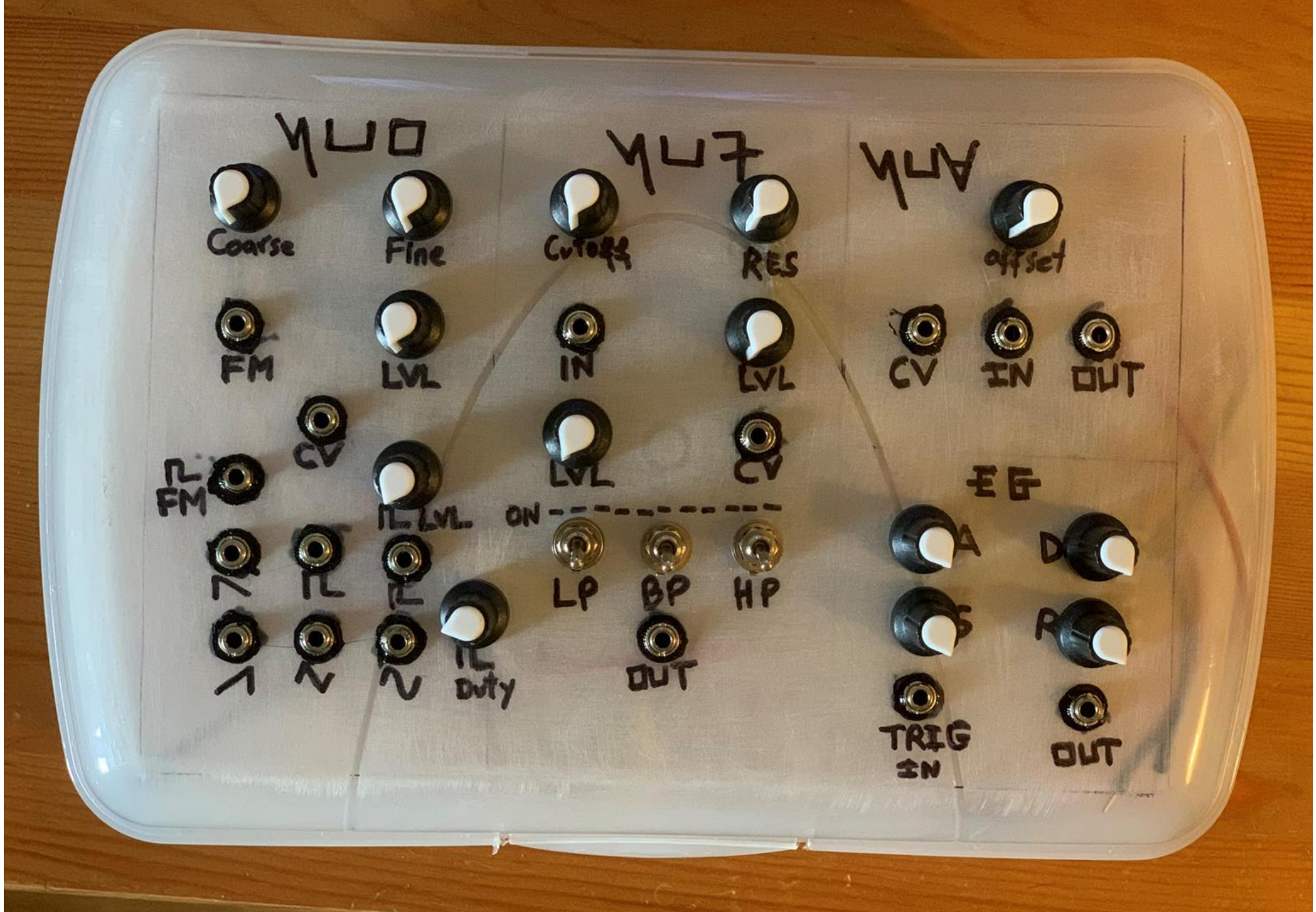
Output Signal
To VCA

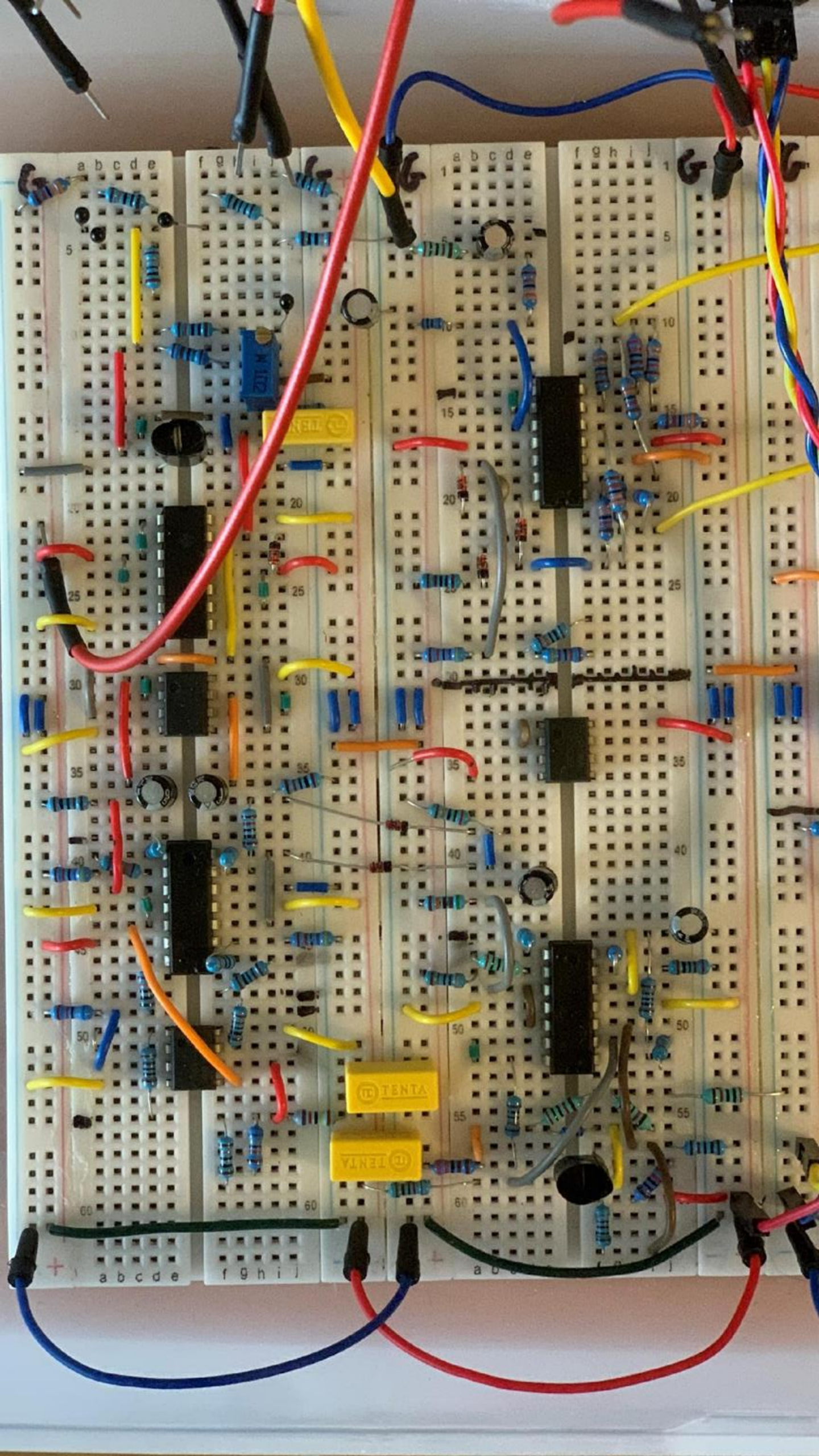


Power Supply


LED
Indicator
+12
V
GN
D
-
12V
Aluminum Foil + thermal
paste
For heat transfer
(Regulators + Transistors)

Results & Evalution


VCO
Coarse
Fine
FM
LVL
CV
FM
LVL
Duty
VCF
Cutoff
RES
IN
LVL
LVL
CV
ON
LP
BP
HP
OUT
VCA
offset
CV
IN
OUT
EG
A
D
S
R
TRIG IN
OUT


TENTA
TENTA

The results of the project were promising, with the synthesizer performing well across all designed modules.

Tuned Synthesizer Across the range of 0-5V.

Low, Band and high pass Filter options alongside the ability to use more than one at the same time to have more interesting combinations.

Fully functioning VCA with Envelope/controller input.

Envelope generator to generate numerous combination and play different types of presets such as , plucks or pads or bass.

Conclusion

“In theory, theory and practice are the same. In practice, they are not.”

- Albert Einstein

“What we know is a drop, what we don't know is an ocean.”

- Isaac Newton

Experience without theory is blind, but theory without experience is mere intellectual play.

No matter how much you learn, there is always room for more.

I have learned and mastered a lot of skills while building my analog synthesizer. It was challenging, hard, not easy… but **Fruitful!**

Thank you for your attention

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25 June 2024